



# SHOMESWARAN A/L MUGUNTHAN

**Address:** Nilai, Negeri Sembilan  
**Phone:** 01127041275  
**Email:** shomezwaran@gmail.com  
**Linkedin:** www.linkedin.com/in/shomeswaran

## SUMMARY

Motivated Computer Science student specializing in Software Development with strong academic performance and hands-on project experience in mobile applications, web development, AI concepts, and IoT-based systems. Skilled in Flutter, Firebase, Java, Python, JavaScript, PHP, SQL, HTML, and CSS. Reliable, fast learner, and passionate about building user-friendly software solutions, solving problems, and improving technical skills.

## EDUCATION

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| <b>Universiti Teknikal Malaysia Melaka (UTeM)</b>                                                                                   | <b>2023 - Present</b> |
| <ul style="list-style-type: none"><li>Bachelor of Computer Science (Software Development) with Honours</li><li>CGPA: 3.89</li></ul> |                       |
| <b>Kolej Matrikulasi Negeri Sembilan (KMNS)</b>                                                                                     | <b>2022 - 2023</b>    |
| <ul style="list-style-type: none"><li>Physical Science Stream</li><li>CGPA: 4.00 MUET: Band 4.0</li></ul>                           |                       |
| <b>Sekolah Menengah Kebangsaan Dato Mohd Said</b>                                                                                   | <b>2017 - 2022</b>    |
| <ul style="list-style-type: none"><li>SIJIL PELAJARAN MALAYSIA (SPM)</li><li>Result: 8A 2B</li></ul>                                |                       |

## SKILLS

- Programming Languages:** Java, Python, JavaScript, TypeScript, Dart, PHP, SQL, C++, HTML, CSS
- Frameworks, Databases & AI:** React.js, React Native, Flutter, Node.js, REST APIs, MySQL, PostgreSQL, Firebase, Supabase, Machine Learning, NumPy, Pandas, Scikit-learn
- Tools, Platforms & IoT:** Git, GitHub, Visual Studio Code, Android Studio, Figma, Canva, XAMPP, Arduino, ESP32, IoT Development
- Development & Soft Skills:** Full-Stack Web Development, Mobile Application Development, API Development, Database Management, UI/UX Design, Testing, Debugging, System Design, Technical Documentation, Problem-Solving, Teamwork, Communication, Analytical Thinking, Time Management

## LANGUAGE PROFICIENCIES

- Bahasa Melayu (Fluent)
- English (Fluent)
- Tamil (Native)

## PROJECTS

- NutriPulse AI — AI-Powered Health and Nutrition Application**  
Flutter | Firebase | Machine Learning | IoT Health Integration (ESP32)
- Designed an AI-powered mobile health application for nutrition monitoring and personalised health recommendations.
  - Integrated food recognition capabilities to identify meals and support automated calorie and nutritional tracking.
  - Incorporated IoT-based health monitoring features for collecting and displaying relevant health information.
  - Developed personalised recommendations based on users' nutrition records, health data, and lifestyle activities.

## **Farm2Screen — Pineapple Marketplace and Information Application (SULAM)**

Flutter | Firebase | Supabase Storage | Mobile Application Development

- Developed a mobile marketplace application connecting pineapple entrepreneurs with customers through dedicated entrepreneur and customer portals.
- Implemented product management, customer ordering, shopping cart, checkout, order history, and order-status tracking features.
- Integrated Firebase Authentication and Firestore for secure user access, real-time data management, and role-based functionality.
- Used Supabase Storage for product image uploads and developed sales analytics, announcements, and daily sales summary features for entrepreneurs.

## **IPETRO VIRS — Visual Inspection Reporting System**

JavaScript | HTML | CSS | SQL |

- Web Application Developed a web-based inspection reporting system based on API 510 inspection requirements.
- Implemented inspection data management and automated report generation with PDF and Microsoft Word export.
- Built separate frontend and backend modules to support structured inspection workflows and database operations.
- Collaborated with a six-member development team and deployed the completed system as a live web application.

## **Retinal Disease Detection System**

Python | Machine Learning | Image Processing

- Developed an image-based machine learning system to support early retinal disease screening.
- Applied image preprocessing and classification techniques to analyse retinal fundus images.
- Implemented a prediction workflow that assists users in identifying possible retinal abnormalities.
- Designed the system to support healthcare-related decision-making through automated medical image analysis.

## **JPJ Management System**

Console Application | File Handling | Vehicle Management

- Developed a console-based vehicle management system supporting vehicle record administration.
- Implemented insert, delete, update, search, and sorting functions for vehicle information.
- Added vehicle maintenance record management and PUSPAKOM inspection-check features.
- Designed a structured menu-based interface to improve data organisation and usability.

## **EXTRACURRICULAR ACTIVITIES**

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### **• Mandarin Language Video Project — Director & Scriptwriter**

Directed the production of a Mandarin-language video and developed the script, while coordinating team members, filming, presentation, pronunciation, and video-editing activities.

### **• Softball — Co-curricular Participant**

Participated in softball training and a small tournament, developing teamwork, discipline, communication, and sportsmanship skills.

## **CERTIFICATIONS & RECOGNITION**

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- Dean's List Award – Achieved the Dean's List for all completed semesters.
- Oracle Database Foundations – Course Participation | Oracle Academy.
- Introduction to Linear Algebra with MATLAB – Course Completion Certificate | MathWorks Training Services.
- Matriculation Academic Excellence – Graduated with a CGPA of 4.00 and MUET Band 4.0.
- Anugerah Murid Cemerlang – Recognised for achieving 8A in SPM 2021 at SMK Dato' Mohd Said.